

## TIANMU SHAN, China

WMO: 584450

Lat: 30.3494N

Lon: 119.4237E

Elev: 4902

StdP: 12.27

Time Zone: 8.00 (E08)

Period: 86-98

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.2       | 14.7 | -22.9                         | 1.9 | 28.0 | -14.5 | 3.0 | 27.6 | 40.3                  | 21.4 | 36.6 | 27.5 | 18.8                  | 340  | 0.668 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.1                    | 77.6            | 70.2 | 75.4 | 69.1 | 73.2 | 67.9 | 71.4                | 75.8 | 70.2 | 74.0 | 69.0 | 72.2 | 5.3                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.0                            | 133.0 | 74.2 | 69.0 | 128.2 | 72.6 | 68.0 | 123.9 | 71.1 | 38.9          | 76.0 | 37.7 | 74.4 | 36.6 | 72.3 | 75.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| 35.5              | 30.4 | 26.6 | DB                         | 5.2 | 81.2               | 3.5 | 2.5                                                | 2.7 | 82.9       | 0.7  | 84.4       | -1.3 | 85.8       | -3.8 | 87.6 | (4) |
|                   |      |      | WB                         | 3.5 | 73.6               | 3.0 | 1.0                                                | 1.3 | 74.3       | -0.4 | 74.9       | -2.1 | 75.5       | -4.3 | 76.2 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 48.9  | 28.8  | 31.1  | 38.3  | 48.7  | 56.3  | 62.3  | 67.9  | 66.5  | 58.7  | 50.4  | 42.4  | 34.6 | (6)  |
|      |                                                                   | DBStd       | 14.94 | 7.82  | 8.82  | 8.53  | 7.67  | 6.48  | 4.34  | 4.56  | 3.85  | 5.69  | 6.59  | 9.13  | 9.04 | (7)  |
|      |                                                                   | HDD50       | 2502  | 656   | 531   | 374   | 112   | 20    | 0     | 0     | 4     | 75    | 253   | 477   |      | (8)  |
|      |                                                                   | HDD65       | 6095  | 1121  | 949   | 829   | 490   | 275   | 101   | 27    | 30    | 202   | 453   | 677   | 941  | (9)  |
|      |                                                                   | CDD50       | 2109  | 0     | 2     | 10    | 72    | 215   | 368   | 554   | 510   | 263   | 88    | 26    | 1    | (10) |
|      |                                                                   | CDD65       | 226   | 0     | 0     | 0     | 0     | 5     | 19    | 116   | 75    | 11    | 0     | 0     | 0    | (11) |
|      |                                                                   | CDH74       | 324   | 0     | 0     | 0     | 0     | 8     | 11    | 217   | 79    | 9     | 0     | 0     | 0    | (12) |
|      |                                                                   | CDH80       | 10    | 0     | 0     | 0     | 0     | 0     | 0     | 8     | 2     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 12.2  | 12.7  | 12.0  | 12.6  | 12.5  | 12.6  | 12.0  | 12.2  | 12.3  | 12.0  | 11.5  | 12.1  | 12.0 | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 66.5  | 2.2   | 3.2   | 4.9   | 5.9   | 7.8   | 10.0  | 7.9   | 8.8   | 7.5   | 4.0   | 2.8   | 1.8  | (15) |
| (16) |                                                                   | PrecMax     | 82.1  | 4.8   | 8.6   | 10.4  | 9.5   | 15.4  | 20.2  | 15.6  | 16.6  | 19.3  | 12.6  | 8.1   | 5.2  | (16) |
| (17) |                                                                   | PrecMin     | 44.3  | 0.0   | 0.7   | 1.8   | 2.7   | 2.9   | 3.3   | 1.3   | 1.4   | 2.0   | 0.0   | 0.1   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 10.4  | 1.2   | 1.7   | 2.0   | 1.8   | 2.9   | 4.1   | 3.7   | 4.1   | 3.9   | 3.2   | 2.0   | 1.3  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 48.8  | 56.5  | 60.4  | 70.5  | 74.7  | 76.0  | 81.2  | 78.8  | 76.8  | 69.1  | 63.1  | 56.1  |      | (19) |
| (20) |                                                                   | 0.4% MCWB   | 40.9  | 48.3  | 54.2  | 57.2  | 63.3  | 66.6  | 71.7  | 71.1  | 70.2  | 59.8  | 53.7  | 43.0  |      | (20) |
| (21) |                                                                   | 2% DB       | 45.7  | 51.3  | 56.7  | 65.9  | 70.8  | 72.7  | 78.5  | 76.2  | 72.3  | 65.0  | 59.5  | 52.1  |      | (21) |
| (22) |                                                                   | 2% MCWB     | 40.2  | 46.0  | 52.5  | 57.5  | 61.8  | 67.2  | 71.1  | 70.1  | 66.4  | 56.7  | 51.6  | 41.5  |      | (22) |
| (23) |                                                                   | 5% DB       | 43.3  | 48.1  | 53.7  | 61.7  | 67.9  | 70.0  | 76.6  | 73.9  | 69.1  | 61.7  | 56.6  | 49.3  |      | (23) |
| (24) |                                                                   | 5% MCWB     | 39.0  | 44.8  | 50.7  | 56.3  | 60.1  | 65.3  | 70.1  | 68.7  | 65.3  | 56.0  | 49.6  | 41.7  |      | (24) |
| (25) |                                                                   | 10% DB      | 40.4  | 44.3  | 51.1  | 59.0  | 64.9  | 67.9  | 74.3  | 71.7  | 66.2  | 58.6  | 53.9  | 46.3  |      | (25) |
| (26) |                                                                   | 10% MCWB    | 35.5  | 40.4  | 48.4  | 55.1  | 58.6  | 64.4  | 69.1  | 67.4  | 64.2  | 53.9  | 48.0  | 39.1  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 45.6  | 51.0  | 57.3  | 62.8  | 67.7  | 70.3  | 73.5  | 72.5  | 71.2  | 63.7  | 59.4  | 50.0  |      | (27) |
| (28) |                                                                   | 0.4% MCDB   | 46.7  | 53.6  | 58.2  | 65.3  | 71.1  | 72.9  | 78.8  | 76.8  | 76.3  | 67.0  | 60.6  | 50.5  |      | (28) |
| (29) |                                                                   | 2% WB       | 43.7  | 48.3  | 54.1  | 60.1  | 64.8  | 68.5  | 71.9  | 70.9  | 68.4  | 60.6  | 56.0  | 46.8  |      | (29) |
| (30) |                                                                   | 2% MCDB     | 44.7  | 49.9  | 55.2  | 61.7  | 67.6  | 70.9  | 76.7  | 74.6  | 71.3  | 62.2  | 57.3  | 48.3  |      | (30) |
| (31) |                                                                   | 5% WB       | 41.0  | 45.6  | 52.0  | 58.3  | 63.0  | 67.1  | 70.8  | 69.6  | 66.5  | 57.8  | 53.3  | 44.2  |      | (31) |
| (32) |                                                                   | 5% MCDB     | 41.7  | 47.5  | 53.2  | 60.3  | 65.1  | 68.8  | 75.1  | 72.7  | 68.2  | 59.3  | 54.8  | 47.5  |      | (32) |
| (33) |                                                                   | 10% WB      | 36.9  | 41.2  | 49.2  | 56.1  | 61.3  | 65.9  | 69.7  | 68.4  | 64.8  | 55.1  | 50.3  | 41.2  |      | (33) |
| (34) |                                                                   | 10% MCDB    | 38.4  | 43.5  | 50.5  | 58.3  | 63.4  | 67.0  | 73.6  | 70.8  | 65.9  | 57.2  | 51.6  | 45.3  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 10.3  | 10.5  | 10.7  | 11.0  | 10.1  | 7.9   | 8.1   | 7.6   | 7.9   | 9.7   | 11.0  | 11.1  |      | (35) |
| (36) |                                                                   | 5% DB MCDBR | 12.8  | 14.2  | 13.7  | 14.2  | 13.3  | 10.9  | 12.2  | 11.2  | 11.2  | 12.6  | 12.6  | 13.4  |      | (36) |
| (37) |                                                                   | 5% DB MCWBR | 9.9   | 10.5  | 10.8  | 10.6  | 10.3  | 8.6   | 7.9   | 7.9   | 9.5   | 9.9   | 9.9   | 10.5  |      | (37) |
| (38) |                                                                   | 5% WB MCDBR | 10.1  | 11.9  | 11.4  | 11.1  | 10.5  | 8.3   | 10.6  | 9.6   | 9.1   | 10.3  | 11.0  | 11.7  |      | (38) |
| (39) |                                                                   | 5% WB MCWBR | 10.2  | 10.1  | 10.0  | 9.3   | 8.6   | 6.6   | 7.6   | 7.3   | 7.7   | 8.4   | 10.0  | 10.1  |      | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.435 | 0.470 | 0.500 | 0.504 | 0.483 | 0.509 | 0.482 | 0.500 | 0.489 | 0.462 | 0.438 | 0.416 |      | (40) |
| (41) |                                                                   | taud        | 1.940 | 1.880 | 1.842 | 1.891 | 2.002 | 1.987 | 2.116 | 2.039 | 2.005 | 1.980 | 1.980 | 1.989 |      | (41) |
| (42) |                                                                   | Ebn at Noon | 241   | 246   | 249   | 254   | 259   | 252   | 258   | 252   | 250   | 246   | 239   | 240   |      | (42) |
| (43) |                                                                   | Edh at Noon | 49    | 57    | 64    | 63    | 57    | 57    | 50    | 54    | 54    | 51    | 47    | 45    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 671   | 775   | 1089  | 1285  | 1395  | 1262  | 1606  | 1418  | 1154  | 1025  | 771   | 689   |      | (44) |
| (45) |                                                                   | RadStd      | 118   | 189   | 95    | 158   | 168   | 190   | 196   | 189   | 118   | 166   | 116   | 132   |      | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|      | Regional (7 neighbors) | +0.58 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +187  | +92   |

Nomenclature: See separate page